

BOB-124 “ Deep Forensic Investigation: 5th Forced-Logout (2026-08-19 15:28:22 CEST)

Revision: 1 Last modified: 2026-08-19T15:45:00Z **Verdict (Â§11.4.6):** SIGKILL INITIATOR **UNATTRIBUTED** “ pattern MATCHES #2/#3/#4 signature for every observable class; kernel-audit attribution still impossible (auditd never installed). **Iron Law:** Investigation only “ no fixes, no source changes.

Executive summary

user@1000.service was SIGKILLED at 2026-08-19 15:28:22 CEST exactly 20 min 41 s after login (15:07:41 â†’ 15:28:22). The kill signature is **identical in shape** to prior incidents #2/#3/#4: a pam_tcb(gdm-password:session): Session closed for milosvasic on tty2 ses=4 fires from gdm-session-worker[3976] at the SAME journal second as the Main process exited, code=killed, status=9/KILL on user@1000.service, followed by a mass SIGKILL cascade of every child under the cgroup. No preceding systemctl stop, loginctl terminate-user, Manager.StopUnit, or KillUnit D-Bus call appears in the journal window (docs/qa/BOB-124/evidence/stop_terminate_paths.log lists only Removed session 1/4 from systemd-logind *after* the SIGKILL). Every kernel-side failure class remains ruled OUT: **no** oom-kill in the kernel journal (evidence/kernel_oom_scan.log empty), **no** systemd-oomd trigger (evidence/oomd_journal.log shows only the boot-time Start), **no** suspend/hibernate, **no** power event. Linger=yes is set (evidence/loginctl_user_state.txt) “ which should preserve user@1000.service across session teardown, yet did not. This is the fifth consecutive incident where Phase 1 root-cause investigation cannot complete because auditd/auditctl/ausearch are not installed and /etc/audit/ does not exist (evidence/loginctl_user_state.txt bottom lines). Correlate signal: user@1000.service: Consumed 1h 21.484s CPU time, 23.6G memory peak, 488.2M memory swap peak in 20 min 41 s “ a memory-accumulation rate roughly Å—10 faster than the prior session (23.6 GB / 21 min vs 25.1 GB / 3 h 22 min reported in docs/incidents/2026-08-19-5th-forced-logout.md), suggesting the reboot did not reset the runaway consumer. The mechanism direction hardens: something INSIDE user@1000 pushes memory into the multi-GB range while sending a gdm-password session-close on tty2, and the SIGKILL falls on the user manager PID milliseconds later “ but no in-scope observer can name the initiator.

Timeline (evidence-cited)

Time (CEST)	Event	Evidence
15:06:20â€“35	Prior session clean shutdown (SERVICE_STOP res=success) â€“ operatorâ€™s intentional reboot	docs/incidents/2026-08-19-5th-forced-logout.md Â§â€œDistinction from the 15:06 clean shutdownâ€“
15:07:39â€“40	Host boot; swap activated (16 GB nvme + 32 GB zram); systemd-oomd.service started	docs/qa/BOB-124/evidence/oom_extended.log lines 3â€“15
15:07:41	user@1000.service started for milosvasic	evidence/user1000_grepped.log lines 1â€“2
15:07:40	milosvasic session on tty2 (ses=4) â€“ see loginctl_user_state.txt Timestamp=Wed 2026-08-19 15:07:40 CEST	evidence/loginctl_user_state.txt
15:07:41 â†’ 15:28:21	20 min 41 s of active work; wave-6 subagents Q/R/S/T dispatched at 15:12 (per .superpowers/sdd/progress.md); scaling_probe pytest hits saturate the merge-search rate limiter (many 429 Too Many Requests in journal_inc5.log)	.superpowers/sdd/progress.md â€œWave-6 in flight â€“ 2026-08-19 15:12â€“; journal_inc5.log 15:28:00-01 batches of ratelimit 10/min exceeded
15:27:14 â†’ 15:28:15	overlayfs xino warnings + crun-buildah scopes: container build activity (BOB-096 tests reload)	evidence/kernel_events.log
15:28:08 â†’ 15:28:19	helix-postgres errors: relation "background_tasks" does not exist (unrelated helix-server churn), then unexpected EOF on client connection with an open transaction	evidence/pre_kill_15s_window.log
15:28:22.181	audit type=1106: op=PAM:session_close ... acct="milosvasic" exe="/usr/libexec/gdm-session-worker" ... terminal=/dev/tty2 res=success	evidence/kill_moment.log lines 6996â€“7002

Time (CEST)	Event	Evidence
15:28:22	from gdm-session-worker[3976] systemd[1]: user@1000.service: Main process exited, code=killed, status=9/ KILL	evidence/kill_moment.log line 7005
15:28:22	Mass SIGKILL cascade "90+ children including postgres, qbittorrent-nox, webui-bridge, tmux: server, claude, python3, MainThread, hook-linux-amd64, lumen-linux-amd, chrome-devtools, caddy, jetbrainsd, yandex_browser" — 15 user@1000.service: Failed with result 'signal' / Consumed 1h 21.484s CPU time, 23.6G memory peak, 488.2M memory swap peak	evidence/systemd_pid1_prekill.log
15:28:23	systemd-logind[1296]: Removed session 1 / Removed session 4 "AFTER the SIGKILL, not before	evidence/user1000_grepped.log bottom
15:28:22	GDM greeter reappears on tty1 (gdm-launch-environment ... session opened for gdm-greeter)	evidence/stop_terminate_paths.log
15:28:35	milosvasic re-authenticated; session-7 (wayland/class=user) + session-8 (manager); user@1000 started again	evidence/stop_terminate_paths.log line 7478; evidence/user1000_grepped.log last two lines

Elapsed dead: 13 s (15:28:22 → 15:28:35 relogin). GDM auto-reset the greeter within 1 s.

Comparative signature table across #2 / #3 / #4 / #5

Sources: docs/incidents/2026-08-18-perceived-forced-logout-2nd.md (#2), docs/incidents/2026-08-18-3rd-forced-logout.md (#3), docs/

incidents/2026-08-19-4th-forced-logout.md + docs/qa/B0B-123/
incident-4-forensics.log (#4), THIS file (#5).

Signature	#2 (2026-08-18 20:50)	#3 (2026-08-18 23:45)	#4 (2026-08-19 00:37)	#5 (2026-08-19 00:37)
gdm-session-worker PAM session_close acct=milosvasic at exact SIGKILL timestamp terminal=/dev/tty2 res=success in the audit1106 user@1000.service: Main process exited, code=killed, status=9/KILL (bare, no preceding Stopping/ Deactivating)	YES (per #4 doc)	YES (per #4 doc)	YES (inc4_forensics_head.txt line 3116)	YES (per #4 doc) L6996
Explicit loginctl terminate / systemctl stop user@1000 / Manager.StopUnit / KillUnit in window	YES	YES	YES	YES
Linger=yes (should have preserved user@1000)	YES	YES	YES	YES (per #4 doc) systemd
IdleAction=ignore (verified in #4)	NO (per #4 doc) "Journal search" returns EMPTY	NO	NO	NO (per #4 doc) stop only per #4 doc 1/4)
Kernel OOM (oom-kill, invoked oom-killer) fired	YES (per #4 doc)	YES	YES (inc4_forensics_head.txt Â§3)	YES (per #4 doc) login
systemd-oomd triggered a kill decision	verified	verified	verified	not recorded " /e not recorded reproducible
Suspend/hibernate/ power-state event	NO	NO	NO	NO (per #4 doc) kernel
KillUserProcesses runtime state	NO	NO	NO	NO (per #4 doc) oomd Start a power lines a keybin transi
	unset (per #4 doc)	unset not recorded	unset (per #4 doc) not recorded in #4 doc	not recorded run th

Signature	#2 (2026-08-18 20:50)	#3 (2026-08-18 23:45)	#4 (2026-08-19 00:37)	#5 (2026-08-19 00:37)
user@1000 memory peak at teardown	not recorded per-incident			23.6 GB (evidence user1 " ac user1
user@1000 CPU peak at teardown	"	"	"	1 h 2 (â%o^
Load average at incident window	not captured	not captured	0.20, 0.70, 1.06 (light)	0.80, (docs, foren
Parallel subagent load at kill	3+ concurrent (per #4 "defensive move")	3+	3+ (Task 85/86/BOB-122/BOB-118 landed)	4 con at 15: pytes
auditctl / ausearch / /etc/ audit/rules.d/ present	NO	NO	NO (per #4 "Recommended next step")	NO (a etc/a
Session dead-time before relogin	"	"	13 min	13 s (

Signature match ratio: every checkable dimension for #5 matches #4 exactly. The two "not re-checked" cells (IdleAction, KillUserProcesses) were verified stable in #4 and no reboot changed logind.conf " but evidence/loginctl_user_state.txt shows IdleHint=no IdleSinceHint=0, i.e. A logind considered the session non-idle at teardown.

Session context (Â§11.4.6, what the session was doing)

Between 15:12 and 15:28 the SDD orchestration dispatched **Wave-6** " four concurrent Claude subagents (Agents Q/R/S/T per .superpowers/sdd/progress.md) plus the merge-search scaling_probe pytest fleet was hammering /api/v1/search (journal_inc5.log at 15:28:00-01 shows " ¥12 different scaling_probe_* query ids being fanned across trackers, tripping the 10/min rate limiter " 429 Too Many Requests bursts). Agent S completed BOB-096 chaos tests at commit 76d6b1f6; other Wave-6 agents were mid-flight. crun-buildah scopes at 15:28:10-15 indicate container build/reload activity. pgrep -af pytest at investigation time returned empty (post-kill), consistent with the SIGKILL cascade having reaped them. This matches the #4 forensic note that "all 4 incidents occurred during heavy parallel subagent load" " #5 makes it 5-of-5.

The load-vs-memory correlation: user@1000 consumed 23.6 GB peak in 20:41 elapsed. The prior session (per docs/incidents/2026-08-19-5th-forced-logout.md) consumed 25.1 GB in 3 h 22 min. The per-minute rate for the killed session is **1.14 GB/min** vs **0.124 GB/min** for the prior — an order of magnitude faster. Whether this rate is causal (systemd-oomd on some ancestor slice, some resource-cap watchdog, or a memory-triggered PAM-close policy) remains **UNCONFIRMED**: no oomd/OOM/pressure event fired, /proc/pressure/memory at investigation time shows avg60=0.00, and the initiator is not self-attributed anywhere in the journal.

What ruled OUT this incident (evidence/cited)

- **Kernel OOM-killer:** evidence/kernel_oom_scan.log returns 0 hits across 15:07-15:29
 - **systemd-oomd:** evidence/oomd_journal.log shows only boot-time Start; no kill decision, no memory-pressure log
 - **Suspend / hibernate / power event:** evidence/power_states.log contains only 2 GSD keybinding-registration warnings; kernel journal shows no PM: suspend / Reached target Sleep
 - **Explicit unit-stop invocation** (loginctl terminate-user, systemctl stop user@1000, KillUnit, Manager.StopUnit): evidence/stop_terminate_paths.log contains only *post-kill* Removed session 1 and Removed session 4 from systemd-logind. No preceding stop RPC.
 - **CONST-033 host power-state transitions:** uptime intact (up 26 min at 15:34 — up 31 min at 15:38), no kernel suspend/hibernate messages
 - **Any process explicitly emitting kill -9 visible in journal:** no signal 9 to pid ... audit trail (audit rules absent — see below)
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The recurring architectural gap (unchanged across #2 — #5)

auditd is not installed on this host: command -v auditctl returns nothing, command -v ausearch returns nothing, /etc/audit/ does not exist, /etc/audit/rules.d/ does not exist. The docs/incidents/2026-08-19-sudo-audit-rules-for-operator.md install script (Rev 3) was authored + committed but never run with sudo. Without those rules, the kernel audit subsystem cannot attribute a signal=9 delivery to a specific initiator PID/UID/cmdline. evidence/user_scope_events_head.log and evidence/kill_moment.log prove the audit *subsystem is running* (type=1106/1113/1104 messages present), but the SYSCALL kill and PATH exec loginctl|systemctl watches that would name the initiator are absent from the rules file that does not exist.

Per docs/incidents/2026-08-19-5th-forced-logout.md
— Systematic-debugging Phase 4.5 — (Iron Law: NO FIXES WITHOUT

ROOT CAUSE INVESTIGATION FIRST + “if % fixes failed: STOP and question the architecture” (the architectural problem is not the SIGKILL mechanism “it is the audit-install gap that has now blocked Phase 1 attribution 5 consecutive times.

What is genuinely NEW in #5 vs #4

1. **Memory-consumption rate** —10 faster than the prior session “ 23.6 GB in 20:41 vs 25.1 GB in 3 h 22 min. The reboot did NOT reset the runaway consumer; whatever pattern of subagent-driven memory growth surfaced in prior sessions surfaced again immediately after a clean boot with the same wave-6 orchestration profile.
2. **Fastest recovery** “ 13 s dead-time vs #4’s 13 min. This suggests operator was actively at the console and re-authenticated immediately.
3. **Container build activity in the pre-kill window** “ crun-buildah scopes at 15:28:10-15 and overlayfs xino warnings across 15:27:14 ‘ 15:28:15 (evidence/kernel_events.log). Not observed being called out for #2/#3/#4 timelines.
4. **Rate-limiter saturation** “ the merge-search proxy issued dozens of 429 Too Many Requests responses in the seconds before the kill, indicating heavy synthesised load from the scaling_probe pytest fleet.

None of these change the SIGKILL initiator attribution “ all four are *correlate signals*, not causal proofs (§11.4.6).

Verdict (§11.4.6)

§11.4.6 Iron-Law status: STILL INCOMPLETE. Every observable class ruled out in #4 is ruled out again in #5. Every signature that fired in #4 fired in #5 in identical shape. The initiator remains unattributed because kernel audit rules are still not installed on the host. #5 matches #2/#3/#4 as one class of incident with high confidence: identical PAM close signature + identical bare SIGKILL exit shape + identical Linger=yes contradiction + identical absence of every candidate mechanism (kernel OOM, systemd-oomd, explicit stop RPC, power event). The load-and-memory correlation is stronger in #5 than in prior incidents but neither #4’s nor #5’s data can distinguish CAUSE from CORRELATION without the audit-rules attribution.

Evidence files (all under docs/qa/B0B-124/)

- journal_inc5.log “ full journalctl 15:25-15:30 (11,330 lines)
- evidence/kill_moment.log “ filtered kill window
- evidence/systemd_pid1_prekill.log “ every systemd[1] message 15:28:15-22
- evidence/pre_kill_15s_window.log “ 15-s pre-kill filtered journal

- evidence/signals_at_kill.log â€” GDM/mutter/logind/PAM signals at kill
- evidence/kernel_events.log â€” kernel messages (overlayfs, veth, rfskill; **no OOM**)
- evidence/kernel_oom_scan.log â€” **empty** (0 kernel-OOM hits)
- evidence/oomb_journal.log â€” systemd-oomd unit journal (Start only, no kill decisions)
- evidence/oom_extended.log â€” extended OOM/PSI/swap scan
- evidence/power_states.log â€” 2 hits (both false-positive keybinding registrations)
- evidence/stop_terminate_paths.log â€” logind session removals (all post-kill)
- evidence/gdm_compositor_events.log â€” GDM/gnome-shell activity around kill
- evidence/gdm_worker_3976.log â€” the actor of the PAM close (PID 3976 = gdm-session-worker)
- evidence/user1000_grepped.log â€” full user@1000.service lifecycle in window
- evidence/user1000_service_journal.log â€” helix-server journal noise (unrelated to kill)
- evidence/user_scope_events_head.log â€” user/session scope timeline (100-line head)
- evidence/loginctl_user_state.txt â€” Linger=yes State=active, Timestamp=15:07:40, IdleHint=no
- evidence/proc_pressure_memory.txt â€” current PSI (avg60=0.00 at investigation time)
- evidence/free_now.txt â€” memory state at investigation
- evidence/inc4_forensics_head.txt â€” first 80 lines of #4 forensics for comparison
- evidence/kernel_oom_scan.log â€” 0 hits, ruling out kernel OOM

Cross-references

- docs/incidents/2026-08-18-perceived-forced-logout-2nd.md (BOB-116, #2)
- docs/incidents/2026-08-18-3rd-forced-logout.md (BOB-120, #3)
- docs/incidents/2026-08-19-4th-forced-logout.md (BOB-123, #4)
- docs/incidents/2026-08-19-5th-forced-logout.md (BOB-124, #5)
- docs/incidents/2026-08-19-sudo-audit-rules-for-operator.md (Rev 3 â€” still uninstalled)
- docs/proposals/external-watchdog-for-forced-logout-architectural-gap.md (Path 2 proposal)
- ~/.claude-claude4/.../memory/forced_logout_incidents.md (playbook)

Iron Law compliance

No source changes made. No sudo invoked. No auditd installed. No shell scripts modified. Only evidence files written under docs/qa/BOB-124/.